

Roughness-Scale Project — Key Equations

1 Saint-Venant Equations with Manning Friction

1.1 Shallow Water Equations (1-D)

Continuity:

$$\frac{\partial h}{\partial t} + \frac{\partial(Uh)}{\partial x} = p - i \quad (1)$$

Momentum:

$$\frac{\partial U}{\partial t} + U \frac{\partial U}{\partial x} + g \frac{\partial h}{\partial x} + g(S_f - S_0) = 0 \quad (2)$$

1.2 Manning's Friction Closure

Friction slope:

$$S_f = \frac{n^2 U^2}{h^{4/3}} \quad (3)$$

Unit discharge (kinematic wave):

$$q = \frac{1}{n} h^{5/3} S_0^{1/2} \quad (4)$$

2 Equivalent Roughness: the Effect Ratio

The *effect ratio* $n_e/\langle n \rangle$ is the equivalent Manning's n normalised by the spatial mean. A value below unity means the heterogeneous surface behaves as if it were smoother than the average roughness.

2.1 Lotter Equation (Equal-Discharge Condition)

The equal-discharge condition requires that total discharge from the heterogeneous surface match that from a uniform surface with n_e :

$$n_e = \frac{\langle h \rangle^{5/3}}{\langle \frac{h^{5/3}}{n} \rangle} \quad (5)$$

Dividing by $\langle n \rangle$ gives the effect ratio in Lotter form:

$$\frac{n_e}{\langle n \rangle} = \frac{\langle h \rangle^{5/3}}{\langle n \rangle \langle \frac{h^{5/3}}{n} \rangle} \quad (6)$$

2.2 Jensen's Inequality

Because $h^{5/3}$ is convex:

$$\langle h^{5/3} \rangle > \langle h \rangle^{5/3} \quad (7)$$

This is the fundamental reason $n_e/\langle n \rangle < 1$: vegetation induces depth variance; the nonlinear weighting $h^{5/3}$ means fast (deep, smooth) zones carry disproportionately more discharge.

3 Spatial Decomposition of S_f

3.1 Reynolds Decomposition

Decompose the spatially varying fields into mean and fluctuation:

$$n = \langle n \rangle + n', \quad U = \langle U \rangle + U', \quad h = \langle h \rangle + h' \quad (8)$$

Normalised fluctuations:

$$r = \frac{n'}{\langle n \rangle}, \quad v = \frac{U'}{\langle U \rangle}, \quad \eta = \frac{h'}{\langle h \rangle} \quad (9)$$

3.2 Second-Order Expansion of Spatially Averaged S_f

$$\frac{\langle S_f \rangle \langle h \rangle^{4/3}}{\langle n \rangle^2 \langle U \rangle^2} \approx 1 + \langle r^2 \rangle + \langle v^2 \rangle + 4\langle rv \rangle - \frac{8}{3}(\langle r\eta \rangle + \langle v\eta \rangle) + \frac{14}{9}\langle \eta^2 \rangle \quad (10)$$

Term	Interpretation
$\langle r^2 \rangle$	Roughness variance
$\langle v^2 \rangle$	Velocity variance (fast/slow zones)
$4\langle rv \rangle$	Roughness–velocity covariance
$-\frac{8}{3}\langle r\eta \rangle$	Depth–roughness covariance
$-\frac{8}{3}\langle v\eta \rangle$	Depth–velocity covariance
$\frac{14}{9}\langle \eta^2 \rangle$	Depth variance

4 Correction Factor: Second-Order Taylor Expansion

The effect ratio expanded to second order in (r, v, η) :

$$\frac{n_e}{\langle n \rangle} \approx 1 - \langle r^2 \rangle - \frac{5}{9}\langle \eta^2 \rangle + \frac{5}{3}\langle r\eta \rangle \quad (11)$$

Alternative (velocity-based):

$$\frac{n_e}{\langle n \rangle} \approx 1 + \frac{1}{4}\langle r^2 \rangle + \frac{5}{4}\langle v^2 \rangle + \frac{5}{2}\langle rv \rangle \quad (12)$$

4.1 Prediction Hierarchy

T_0 Leading term: uses only roughness variance $\langle r^2 \rangle$ (known from the n -field alone).

T_1 Adds roughness–velocity and roughness–depth covariances.

T_2 Full 2nd-order: all six terms — best accuracy but requires simulation fields.

CF approx Replaces simulation covariances with analytical correction factors.

5 Composite Channel Equations

Classical equivalent-roughness formulas designed for compound channels, applied here to patchy hillslopes. Each formula computes a composite roughness n_c from the spatially distributed Manning's n_i and (where needed) depth h_i .

#	Method	Formula
1	Lotter	$n_c = \frac{\bar{h}^{5/3}}{\left\langle \frac{h_i^{5/3}}{n_i} \right\rangle}$
2	Pavlovskii	$n_c = \sqrt{\langle n_i^2 \rangle}$
3	Horton–Einstein	$n_c = \langle n_i^{3/2} \rangle^{2/3}$
4	Cox (arith. mean)	$n_c = \langle n_i \rangle$
5	Krishnamurthy–Chr.	$n_c = \exp \left(\frac{\langle h_i^{3/2} \ln n_i \rangle}{\langle h_i^{3/2} \rangle} \right)$
6	Colebatch	$n_c = \left(\frac{\langle h_i n_i^{3/2} \rangle}{\bar{h}} \right)^{2/3}$
7	Yen $R^{1/6}$	$n_c = \bar{h}^{1/6} \left\langle \frac{n_i}{h_i^{1/6}} \right\rangle$
8	Yen $R^{1/3}$	$n_c = \frac{\langle h_i^{1/3} n_i \rangle}{\bar{h}^{1/3}}$
9	Felkel	$n_c = \frac{1}{\langle n_i^{-1} \rangle}$

Table 1: Composite-channel equations for equivalent roughness. n_i = Manning's coefficient at cell i ; h_i = depth at cell i ; $\bar{h} = \langle h_i \rangle$ = spatial mean depth; $\langle \cdot \rangle$ = spatial average.

5.1 Lotter Reduces to Unity Under CF Approximation

Under the kinematic-wave correction-factor (CF) approximation, $h_i \propto n_i^{3/5}$, so

$$h_i^{5/3} = C \cdot n_i \quad \text{where } C = (q_0 l / S_0^{1/2})$$

is cell-independent. Therefore $\langle h^{5/3} / n \rangle = C$ and $n_c = \bar{h}^{5/3} / (C \langle n \rangle)$, which equals $\langle n \rangle$, making $n_c / \langle n \rangle = 1$ — i.e., the Lotter equation becomes uninformative when depth variations follow the kinematic-wave scaling.

6 Length-Scale Attribution of S_f Decomposition

The six second-order terms T_k from (10) sum to a departure score

$$S = \sum_{k=1}^6 T_k = \langle r^2 \rangle + \langle v^2 \rangle + 4\langle rv \rangle - \frac{8}{3}(\langle r\eta \rangle + \langle v\eta \rangle) + \frac{14}{9}\langle \eta^2 \rangle \quad (13)$$

that controls how far the effect ratio departs from unity. When S varies across patch lengthscales (L_V), the following methods identify which terms drive that variation.

6.1 Covariance-Based Variance Attribution

The key question is: *when we vary the vegetation lengthscale L_V across simulations, which of the six terms T_k are responsible for the resulting spread in S ?*

Since $S = \sum_k T_k$, a standard identity from probability theory gives an exact decomposition of variance:

$$\text{Var}(S) = \sum_{k=1}^6 \text{Cov}(T_k, S) \quad (14)$$

Dividing both sides by $\text{Var}(S)$ yields the *attribution fraction* for term k :

$$\alpha_k = \frac{\text{Cov}(T_k, S)}{\text{Var}(S)}, \quad \sum_{k=1}^6 \alpha_k = 1 \quad (15)$$

Interpretation of sign.

- $\alpha_k > 0$: term T_k co-varies with S in the same direction — it *drives* the cross-scale variability. When S is large, T_k also tends to be large.
- $\alpha_k < 0$: term T_k moves opposite to S — it *dampens* or partially offsets the other terms.
- The fractions always sum to exactly 1 by construction, regardless of sample size.

The cross-terms with negative coefficients in S (namely $-\frac{8}{3}\langle r\eta \rangle$ and $-\frac{8}{3}\langle v\eta \rangle$) are the most likely candidates for negative α_k , since their weighted products naturally tend to oppose the sum. The pure variance terms ($\langle r^2 \rangle$, $\langle \eta^2 \rangle$) tend to stay positive because variances are non-negative and generally increase with L_V .

Grouping. The covariance in (14) is computed across the set of simulations that differ in L_V within a fixed group. Two grouping strategies are used:

1. **By anisotropy:** gradient-aligned, isotropic, and contour-aligned, with f_V and storm forcing pooled within each group.
2. **By vegetation fraction:** f_V is binned into three equal-frequency classes (via quantile binning) to ensure sufficient sample sizes. Anisotropy and storm forcing are pooled within each bin.

Stability considerations. When the number of simulations per group is small ($n \lesssim 5$), the covariance estimates can be noisy and individual α_k values may exhibit spurious sign flips — particularly for the cross-terms whose true values are close to zero. Pooling across all converged storms (rather than conditioning on a single storm event) and binning f_V into terciles rather than using raw levels substantially improves stability.

6.2 Slope Decomposition (Fixed-Effect Regression)

While the covariance attribution answers “what fraction of spread is due to each term?”, the slope decomposition asks a more direct question: *how much does each T_k change per unit increase in L_V ?*

Fit each term with an OLS model that includes L_V as the predictor of interest and dummy variables to absorb group-level variation:

$$T_k = \beta_{0,k} + \beta_{L,k} L_V + \sum_j \gamma_{j,k} \mathbb{K}_{f_V=j} + \sum_m \delta_{m,k} \mathbb{K}_{\text{storm}=m} + \varepsilon_k \quad (16)$$

where $\mathbb{K}_{f_V=j}$ are vegetation-fraction fixed effects, $\mathbb{K}_{\text{storm}=m}$ are storm-forcing fixed effects (encoding each $p \times t_r$ combination), and $\beta_{L,k}$ is the marginal effect of lengthscale on term k , holding all else constant.

The slopes decompose the total sensitivity of S to patch scale:

$$\frac{\partial S}{\partial L_V} = \sum_{k=1}^6 \beta_{L,k} \quad (17)$$

Terms with large $|\beta_{L,k}|$ are the primary drivers of the scale dependence of S . Standard errors from OLS provide confidence intervals that are unavailable from the model-free covariance attribution.

When the regression is run separately per group (e.g. per anisotropy level or per f_V bin), the fixed effects absorb only the remaining nuisance variables — for instance, when grouping by anisotropy, the design matrix includes f_V and storm fixed effects but not anisotropy dummies.

6.3 Summary: Two Complementary Views

Analysis	Question	Method
Covariance attribution	Which terms explain the <i>spread</i> in S ?	Variance identity (exact, model-free)
Slope decomposition	How much does each term change per metre of L_V ?	OLS regression (gives standard errors)